

Literature Status for Selected Tingley Problems in arXiv:1712.09192

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Status

This is a lightweight literature-implied answer packet. It records that two parts of the problem list in Mori’s paper “Tingley’s problem through the facial structure of operator algebras” (arXiv:1712.09192) are answered by later or parallel literature:

- the reduction problem for unital C^* -algebras is answered by Mori–Ozawa, arXiv:1804.10674;
- the Schatten-class subcase of the noncommutative L^p problem is answered by Fernandez–Polo–Jorda–Peralta, arXiv:1803.00763.

The general self-adjoint part problem for unital C^* -algebras and the full Haagerup $L^p(M)$ problem for arbitrary von Neumann algebras are not settled by this packet.

Source Problems

In Section 6 of arXiv:1712.09192, the source asks:

Let A be a unital C^* -algebra and $\Phi : S(A) \rightarrow S(A)$ be a surjective isometry. Suppose that $\Phi(x) = x$ for every $x \in S(A) \cap A^{-1}$. Is Φ equal to the identity mapping on $S(A)$?

The same section asks Tingley’s problem for the spaces A_{sa} of self-adjoint elements in unital C^* -algebras, and for Haagerup noncommutative L^p -spaces $L^p(M)$, $1 < p < \infty$, $p \neq 2$.

Supporting Literature

Mori–Ozawa, “Mankiewicz’s theorem and the Mazur–Ulam property for C^* -algebras” (arXiv:1804.10674), prove as Theorem **thm:mau** that every unital complex C^* -algebra, regarded as a real Banach space, has the Mazur–Ulam property. Hence any surjective sphere isometry $\Phi : S(A) \rightarrow S(A)$ extends to a real linear surjective isometry $L : A \rightarrow A$. If Φ fixes all invertible norm-one elements, then L fixes every unitary. By the Russo–Dye theorem, the closed convex hull of the unitary group is the closed unit ball, so L is the identity on the ball and therefore $\Phi = \text{id}$ on the sphere. This answers the source’s **problem-C***; the source itself notes after submission that Mori and Ozawa announced results containing an affirmative solution.

Fernandez–Polo–Jorda–Peralta, “Tingley’s problem for p -Schatten von Neumann classes” (arXiv:1803.00763), state in the abstract that for complex Hilbert spaces H, H' and $p \in (1, \infty) \setminus \{2\}$ every surjective isometry

$$\Delta : S(C_p(H)) \rightarrow S(C_p(H'))$$

extends to a complex linear or conjugate linear surjective isometry between the Schatten classes. This is the $M = B(H)$, standard trace, Schatten-class subcase of the source's noncommutative L^p problem.

Scope and Search Evidence

Cheap run-index search found nearby Tingley/Mazur–Ulam packets but no existing packet for arXiv:1712.09192. Local source search found arXiv:1804.10674 and arXiv:1803.00763 as decisive supporting papers. Web search on 2026-06-28 for the source title, Mori–Ozawa C^* MUP, self-adjoint C^* Tingley, and noncommutative L^p Tingley confirmed the source and the Mori–Ozawa follow-up but did not reveal a full later solution of the remaining two general problems.